

Reverse-check review package — Wprime_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	Wprime/model/Wprime_gen.fr
original model name	Wprime_gen (hidden from the agent)
paper	Wprime/text/hep-ph_0207290.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LkinWp (=)

```
Block[{mu,nu}, -1/2 DC[HC[Wp[nu]],mu] DC[Wp[nu],mu] + 1/2 DC[HC[Wp[nu]],mu] DC[Wp[mu],nu] +  
  ↪ MWp^2 HC[Wp[mu]] Wp[mu]]
```

LkinNuR (=)

```
Block[{mu,sp1,sp2,i}, Sum[I bar[vR][sp1,i].Ga[mu,sp1,sp2].del[vR[sp2,i],mu] - MvR  
  ↪ bar[vR][sp1,i].vR[sp1,i], {i,1,3}]]
```

LWpQuarkNonHC (:=)

```
Block[{mu,sp1,sp2,sp3,i,j,aa}, Sum[1/Sqrt[2] Wp[mu] (gWpR Exp[I omegaWp] Cos[zetaWp]  
  ↪ VpRQ[i,j] bar[uq][sp1,i,aa].Ga[mu,sp1,sp2].ProjP[sp2,sp3].dq[sp3,j,aa] + gWpL  
  ↪ Sin[zetaWp] VpLQ[i,j] bar[uq][sp1,i,aa].Ga[mu,sp1,sp2].ProjM[sp2,sp3].dq[sp3,j,aa]),  
  ↪ {i,1,3}, {j,1,3}, {aa,1,3}]]
```

LWpLeptonNonHC (:=)

```
Block[{mu,sp1,sp2,sp3,i,j}, Sum[1/Sqrt[2] Wp[mu] (gWpR Exp[I omegaWp] Cos[zetaWp] VpRL[i,j]  
  ↪ bar[vR][sp1,i].Ga[mu,sp1,sp2].ProjP[sp2,sp3].l[sp3,j] + gWpL Sin[zetaWp] VpLL[i,j]  
  ↪ bar[vL][sp1,i].Ga[mu,sp1,sp2].ProjM[sp2,sp3].l[sp3,j]), {i,1,3}, {j,1,3}]]
```

LWpFermions (:=)

```
LWpQuarkNonHC + LWpLeptonNonHC + HC[LWpQuarkNonHC + LWpLeptonNonHC]
```

LBSM (=)

```
LkinWp + LkinNuR + LWpFermions
```

Blank-slate reconstruction

Reconstructed Physics From `sanitized.fr`

Lagrangian

The model defines

$$\mathcal{L}_{\text{BSM}} = \mathcal{L}_{\text{LkinWp}} + \mathcal{L}_{\text{LkinNuR}} + \mathcal{L}_{\text{LWpFermions}}.$$

The new charged vector is denoted $W_\mu'^+ \equiv W_{p\mu}$, with $W_\mu'^- \equiv (W_\mu'^+)^\dagger$.

The file declares only electric charge for W' , $Q(W'^+) = +1$, and no color or weak-isospin indices. Thus the covariant derivative acting on the new vector contains the gauge connection for its declared electric charge and no declared $SU(3)_c$ or $SU(2)_L$ action:

$$D_\mu W_\nu'^+ = (\partial_\mu - ieA_\mu) W_\nu'^+, \quad D_\mu W_\nu'^- = (\partial_\mu + ieA_\mu) W_\nu'^-.$$

Equivalently,

$$W_{\mu\nu}'^+ = D_\mu W_\nu'^+ - D_\nu W_\mu'^+, \quad W_{\mu\nu}'^- = D_\mu W_\nu'^- - D_\nu W_\mu'^-.$$

LkinWp

$$\mathcal{L}_{\text{LkinWp}} = -\frac{1}{2} (D_\mu W_\nu'^-) (D^\mu W'^{+\nu}) + \frac{1}{2} (D_\mu W_\nu'^-) (D^\nu W'^{+\mu}) + M_{W'}^2 W_\mu'^- W'^{+\mu}$$

Equivalently, up to the usual integration-by-parts form for a complex Proca field,

$$\mathcal{L}_{\text{LkinWp}} = -\frac{1}{2} W_{\mu\nu}'^- W'^{+\mu\nu} + M_{W'}^2 W_\mu'^- W'^{+\mu}.$$

LkinNuR

The model introduces three non-self-conjugate right-handed neutrino fields ν_{Ri} , $i = 1, 2, 3$.

$$\mathcal{L}_{\text{LkinNuR}} = \sum_{i=1}^3 \left[i \bar{\nu}_{Ri} \gamma^\mu \partial_\mu \nu_{Ri} - M_{\nu_R} \bar{\nu}_{Ri} \nu_{Ri} \right]$$

This is a Dirac-type mass term as written in the file, not a Majorana term.

LWpQuarkNonHC

With $P_R = (1 + \gamma^5)/2$, $P_L = (1 - \gamma^5)/2$, generation indices $i, j = 1, 2, 3$, and color index $a = 1, 2, 3$,

$$\mathcal{L}_{\text{LWpQuarkNonHC}} = \frac{1}{\sqrt{2}} \sum_{i,j=1}^3 \sum_{a=1}^3 W_\mu'^+ \left[g_{WpR} e^{i\omega_{Wp}} \cos \zeta_{Wp} (V^{R,q})_{ij} \bar{u}_i^a \gamma^\mu P_R d_j^a + g_{WpL} \sin \zeta_{Wp} (V^{L,q})_{ij} \bar{u}_i^a \gamma^\mu P_L d_j^a \right]$$

The color contraction is diagonal, δ_{ab} .

LWpLeptonNonHC

$$\mathcal{L}_{\text{LWpLeptonNonHC}} = \frac{1}{\sqrt{2}} \sum_{i,j=1}^3 W_{\mu}^{\prime+} [g_{WpR} e^{i\omega_{Wp}} \cos \zeta_{Wp} (V^{R,\ell})_{ij} \bar{\nu}_{Ri} \gamma^{\mu} P_R \ell_j + g_{WpL} \sin \zeta_{Wp} (V^{L,\ell})_{ij} \bar{\nu}_i \gamma^{\mu} P_L \ell_j]$$

Here $\nu_i \equiv \nu_{Li}$ is the Standard Model left-neutrino field appearing in the base model, while $\nu_{Ri} \equiv \nu_{Ri}$ is the new right-handed neutrino field declared in this file.

LWpFermions

The full charged-current interaction includes the Hermitian conjugate:

$$\mathcal{L}_{\text{LWpFermions}} = \mathcal{L}_{\text{LWpQuarkNonHC}} + \mathcal{L}_{\text{LWpLeptonNonHC}} + \text{h.c.}$$

Explicitly, the Hermitian conjugate contains $W_{\mu}^{\prime-}$ interactions such as

$$\frac{1}{\sqrt{2}} W_{\mu}^{\prime-} g_{WpR} e^{-i\omega_{Wp}} \cos \zeta_{Wp} (V_{ij}^{R,q})^* \bar{d}_j^a \gamma^{\mu} P_R u_i^a$$

and analogous conjugates for the left-handed quark, right-handed lepton, and left-handed lepton currents.

Field Table

.fr class	Particle symbol	Spin	$SU(3)_c$ rep	$SU(2)_L$ rep	$U(1)$ charge / hypercharge	Self-conjugate?	Mass
Wp	$W_{\mu}^{\prime+}$, antiparticle $W_{\mu}^{\prime-}$	1 vector	singlet, no color index declared	not declared; no weak- isospin index	electric charge $Q = +1$; hy- percharge not declared	no	MWp = 1000.
vR	ν_{Ri} , $i = 1, 2, 3$: vR, v _m R, v _t R	1/2 fermion	singlet, no color index declared	not declared; no weak- isospin index	electric charge $Q = 0$; hy- percharge not declared	no	MvR = 100.

The vector width is declared as `WWp = 1.`. The right-handed neutrino width is declared as zero.

Parameters

Parameter	Value	Multiplies	Physical meaning
gWpL	0.653	Left-chiral W' currents $\bar{u} \gamma^{\mu} P_L d$ and $\bar{\nu} \gamma^{\mu} P_L \ell$, with overall factor $\sin \zeta_{Wp}$	Left-handed charged-current coupling of W'

Parameter	Value	Multiplies	Physical meaning
gWpR	0.653	Right-chiral W' currents $\bar{u}\gamma^\mu P_R d$ and $\bar{\nu}_R\gamma^\mu P_R \ell$, with phase $e^{i\omega_{Wp}}$ and factor $\cos\zeta_{Wp}$	Right-handed charged-current coupling of W'
zetaWp	0.	Appears as $\cos\zeta_{Wp}$ on right-handed currents and $\sin\zeta_{Wp}$ on left-handed currents	Mixing angle controlling right- versus left-chiral W' couplings
omegaWp	0.	Appears as $e^{i\omega_{Wp}}$ multiplying right-handed currents	CP phase in the right-handed W' coupling
VpLQ[i, j]	CKM-like real matrix	Multiplies $\bar{u}_i\gamma^\mu P_L d_j$	Left-handed quark flavor mixing matrix
VpRQ[i, j]	CKM-like real matrix	Multiplies $\bar{u}_i\gamma^\mu P_R d_j$	Right-handed quark flavor mixing matrix
VpLL[i, j]	identity matrix	Multiplies $\bar{\nu}_i\gamma^\mu P_L \ell_j$	Left-handed lepton flavor mixing matrix
VpRL[i, j]	identity matrix	Multiplies $\bar{\nu}_{Ri}\gamma^\mu P_R \ell_j$	Right-handed lepton flavor mixing matrix

The quark mixing matrices are initialized as

$$V^{L,q} = V^{R,q} = \begin{pmatrix} 0.9751 & 0.2215 & 0.0035 \\ 0.2210 & 0.9743 & 0.0410 \\ 0.0090 & 0.0400 & 1.0000 \end{pmatrix}.$$

The lepton mixing matrices are initialized as

$$V^{L,\ell} = V^{R,\ell} = \mathbf{1}_{3\times 3}.$$

Physics Summary

This file encodes a charged massive vector boson $W'^{\pm}$ and three non-self-conjugate right-handed neutrinos ν_{Ri} . The W' couples to charged quark and lepton currents with independently weighted right- and left-chiral structures, controlled by $g_{WpR}\cos\zeta_{Wp}$, $g_{WpL}\sin\zeta_{Wp}$, flavor matrices, and a right-handed CP phase $e^{i\omega_{Wp}}$.

The model mediates charged-current processes such as $q\bar{q}' \rightarrow W'^{\pm}$, $W'^+ \rightarrow u_i\bar{d}_j$, $W'^+ \rightarrow \nu_{Ri}\ell_j^+$, and the Hermitian-conjugate W'^- decays.

Paper cross-check

Comparison of reconstruction.md Against the Paper Model

Paper Lagrangian / Field Definitions Located

The paper defines the model interaction in the Introduction. The primary model-defining Lagrangian is Eq. (1), described as “the most general Lorentz invariant Lagrangian describing the coupling of a W' to fermions”:

$$\mathcal{L} = \frac{g}{\sqrt{2}} \bar{f}_i \gamma^\mu \left(C_{f_i f_j}^R P_R + C_{f_i f_j}^L P_L \right) W'_\mu f_j + \text{H.c.} \quad (1)$$

with $P_{R,L} = (1 \pm \gamma^5)/2$, g the SM $SU(2)_L$ coupling, and $C_{f_i f_j}^{R,L}$ arbitrary quark/lepton-dependent couplings.

The paper then rewrites this in left-right-symmetric notation in Eq. (2):

$$\mathcal{L} = \frac{1}{\sqrt{2}} \bar{f}_i \gamma^\mu \left(g_R e^{i\omega} \cos \zeta V_{f_i f_j}^R P_R + g_L \sin \zeta V_{f_i f_j}^L P_L \right) W'_\mu f_j + \text{H.c.} \quad (2)$$

where ζ is a left-right mixing angle, ω is a CP phase absorbable into V^R , $g_{R,L}$ are right/left gauge couplings, and $V^{R,L}$ are generalized CKM matrices.

Additional relevant definitions appear in Sec. II: the width is decomposed in Eq. (3), leading-order partial widths are Eqs. (4)-(6), the effective coupling combination is Eq. (7),

$$|gV'_{f_i f_j}|^2 = |g_L \sin \zeta V_{f_i f_j}^L|^2 + |g_R \cos \zeta V_{f_i f_j}^R|^2, \quad (7)$$

and the numerical CKM-like matrix is Eq. (24). The right-handed neutrino is discussed in the Introduction and Sec. II as a possible leptonic decay product: a right-handed W'_R decays leptonically only if $m_{\nu_R} < m_{W'}$, or if there is large left-right mixing in the neutrino sector.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
General W' -fermion interaction, $\frac{g}{\sqrt{2}} \bar{f}_i \gamma^\mu (C^R P_R + C^L P_L) W'_\mu f_j + \text{H.c.}$ (Eq. 1)	$\mathcal{L}_{\text{LWP Fermions}} = \mathcal{L}_{\text{LWP Quark NonHC}} + \mathcal{L}_{\text{LWP Lepton NonHC}} + \text{h.c.}$	agree	The reconstruction implements the same charged-current Lorentz structure with P_R and P_L , plus Hermitian conjugate. It is expressed in the more specialized Eq. (2) parameterization rather than arbitrary $C^{R,L}$.
Left-right form $g_R e^{i\omega} \cos \zeta V^R P_R + g_L \sin \zeta V^L P_L$ (Eq. 2)	$g_{WpR} e^{i\omega_{Wp}} \cos \zeta_{Wp} V^R P_R + g_{WpL} \sin \zeta_{Wp} V^L P_L$	agree	Coefficient structure, CP phase placement, mixing-angle factors, and chiral projectors match Eq. (2).
Quark charged current implied by Eqs. (1)-(2), with GCKM matrices for quarks	$W_\mu^{'+} \bar{u}_i^a \gamma^\mu P_{R,L} d_j^a$ with $V_{ij}^{R,q}, V_{ij}^{L,q}$, plus h.c.	agree	This is the expected explicit charged-current orientation for W'^{+} ; color is diagonally contracted, consistent with a color-singlet W' .
Lepton charged current implied by Eqs. (1)-(2), with diagonal lepton couplings in the SM-like case	$W_\mu^{'+} \bar{\nu}_{Ri} \gamma^\mu P_R \ell_j$ and $W_\mu^{'+} \bar{\nu}_i \gamma^\mu P_L \ell_j$, plus h.c.	agree	The reconstruction separates SM left neutrinos and new right-handed neutrinos. This matches the paper's distinction between W'_L leptonic decays and $W'_R \rightarrow \ell \nu_R$ when kinematically allowed.

paper term (eq. ref)	reconstruction term	verdict	notes
Hermitian conjugate in Eqs. (1)-(2)	Explicit h.c. containing $W_{\mu}^{\prime -} \bar{d}_j \gamma^{\mu} P_{R,L} u_i$, etc.	agree	The reconstruction correctly includes charge-conjugate interactions with complex-conjugated mixing matrices and $e^{-i\omega}$.
Projector convention $P_{R,L} = (1 \pm \gamma^5)/2$ (Eq. 1 text)	$P_R = (1 + \gamma^5)/2$, $P_L = (1 - \gamma^5)/2$	agree	Same convention.
Effective width coupling $ gV' ^2 = g_L \sin \zeta V^L ^2 + g_R \cos \zeta V^R ^2$ (Eq. 7)	Separate left/right amplitudes with the same $g_{WpL,R}$, $\sin \zeta_{Wp}$, $\cos \zeta_{Wp}$, and $V^{L,R}$ factors	agree	The reconstruction's Lagrangian is the pre-factorized form that yields the paper's Eq. (7) under the paper's assumption of at most one nonzero final-state mass.
Numerical quark mixing matrix V'_{CKM} (Eq. 24)	$V^{L,q} = V^{R,q} = \begin{pmatrix} 0.9751 & 0.2215 & 0.0035 \\ 0.2210 & 0.9743 & 0.0410 \\ 0.0090 & 0.0400 & 1.0000 \end{pmatrix}$	agree	The reconstructed quark matrices match the paper's numerical matrix.
Lepton mixing assumed identity in numerical results (Sec. II.A, before Eq. 24)	$V^{L,\ell} = V^{R,\ell} = \mathbf{1}$	agree	Matches the paper's numerical assumption $V'_{\ell\nu}$ is the identity.
W' as charged massive vector boson, with both W'^+ and W'^- through h.c. (Introduction, Eqs. 1-2)	Non-self-conjugate vector $W_{\mu}^{\prime +}$, antiparticle $W_{\mu}^{\prime -}$, electric charge $Q = +1$	agree	The explicit charge assignment is consistent with the charged-current interactions and Hermitian conjugate in the paper.
Right-handed neutrino as possible W'_R leptonic decay product, with kinematic condition $m_{\nu_R} < m_{W'}$ (Introduction; Sec. II after Table I)	Three non-self-conjugate ν_{Ri} fields with mass $M_{\nu_R} = 100$	agree	The presence of ν_R agrees. The paper does not specify three generations or a Dirac-type kinetic/mass term, but three flavors are a natural implementation of the lepton matrix structure.
Paper's general arbitrary couplings $C_{f_i f_j}^{R,L}$, allowed to differ for quarks and leptons (Eq. 1 text)	Universal g_{WpR} , g_{WpL} multiplying separate quark/lepton flavor matrices	disagree	The reconstruction matches the left-right Eq. (2) parameterization, but is less general than Eq. (1) if Eq. (1) is taken literally as independent arbitrary $C^{R,L}$ for each fermion sector and flavor entry.

paper term (eq. ref)	reconstruction term	verdict	notes
Paper does not define a standalone W' kinetic or electromagnetic covariant derivative term	$\mathcal{L}_{\text{KinWp}} = -\frac{1}{2}W_{\mu\nu}'W'^{+\mu\nu} + M_{W'}^2 W_{\mu}'W'^{+\mu}$, with EM-only covariant derivative	extra-in-reconstruction	This is necessary for an implementation, but it is not specified by the paper's model definition. The EM-only covariant derivative is an implementation choice not stated in the paper.
Paper does not define a standalone ν_R kinetic or mass Lagrangian	$\mathcal{L}_{\text{KinNuR}} = \sum_i i\bar{\nu}_{Ri}\gamma^\mu\partial_\mu\nu_{Ri} - M_{\nu_R}\bar{\nu}_{Ri}\nu_{Ri}$	extra-in-reconstruction	The paper only refers to the mass of ν_R for whether the decay is open; it does not specify Dirac versus Majorana structure or the number of explicit ν_R fields.
Paper numerical treatment uses calculated W' widths from partial widths, Eqs. (3)-(7), (22)-(23), Tables II-III	Reconstruction declares $\text{Wp} = 1$.	disagree	A fixed width of 1 GeV is not the paper's width prescription. It may be a placeholder implementation parameter, but it does not reproduce the paper's mass- and coupling-dependent total width.
Paper treats g_R , g_L , ζ , ω , and GCKM matrices as model parameters; numerical examples use SM-like coupling normalization	Reconstruction initializes $g_{WpL} = g_{WpR} = 0.653$, $\zeta_{Wp} = 0$, $\omega_{Wp} = 0$	agree	The symbols and coupling normalization are consistent with the paper's SM-like examples. The default $\zeta = 0$ selects a purely right-handed benchmark, not the whole arbitrary-coupling setup.
Paper defines no $SU(2)_L$, hypercharge, or color representation for W' beyond its charged-current role	Reconstruction declares W' with no color or weak-isospin index and electric charge only	agree	This is compatible with the paper's phenomenological treatment, though the paper discusses possible UV origins such as broken $SU(2)_L \times SU(2)_R$.

Disagreements and Checks

disagreement	severity	what a human should check
Reconstruction implements Eq. (2)'s factorized left-right form, while Eq. (1) allows fully arbitrary $C_{f_i f_j}^{R,L}$ couplings that can differ by sector and flavor.	convention	Check whether the implementation is intended to reproduce the paper's Eq. (2) benchmark parameterization or the maximally general Eq. (1) coupling freedom.

disagreement	severity	what a human should check
Reconstruction includes an explicit W' kinetic/mass term with an EM-only covariant derivative, which the paper does not specify.	substantive	Check whether photon interactions of W' generated by the EM covariant derivative are intended and whether they are used in processes beyond the paper's s -channel charged-current calculation.
Reconstruction includes three Dirac-type right-handed neutrino fields with explicit kinetic and mass terms, while the paper only uses m_{ν_R} as a kinematic condition for W'_R leptonic decays.	substantive	Check whether the implementation's Dirac, non-self-conjugate ν_R choice matches the target phenomenology, especially if neutrino-sector mixing or Majorana assumptions matter.
Reconstruction declares a fixed W' width $\mathbf{Wp} = 1.$, whereas the paper computes total widths from the coupling-dependent partial widths in Sec. II.	substantive	Check whether the width is recalculated elsewhere at runtime; if not, the implementation will not reproduce the paper's width and branching-ratio predictions.
Reconstruction default $\zeta_{Wp} = 0$ makes the default interaction purely right-handed, while the paper treats arbitrary left/right couplings and discusses both left- and right-handed cases.	convention	Check whether parameter cards used for comparisons set ζ and couplings to the desired paper scenario rather than relying on the default.

Overall Assessment

The reconstructed charged-current interaction agrees closely with the paper's Eq. (2) left-right-symmetric parameterization: the chiral projectors, $g_R e^{i\omega} \cos \zeta$ and $g_L \sin \zeta$ coefficient structure, Hermitian conjugation, quark CKM-like matrix, and identity lepton mixing all match the paper's definitions and numerical assumptions. The main caveats are that the reconstruction is an implementation-level model rather than only the paper's phenomenological interaction: it adds kinetic and mass terms for W' and ν_R , fixes a default right-handed benchmark, and declares a fixed W' width that does not by itself reproduce the paper's calculated width prescription.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 9 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: [] approve [] approve with corrections [] reject
- Notes: